a
Known signed connectivity
Network J5000: columns presynaptic, rows postsynaptic
N0
Postsynaptic node
Presynaptic node
0
3
6
9
11
N4
N6
N8
N11
−0.42
0.00
+0.42
Connection weight (a.u.): negative to positive
recorded nodes N6, N8, N11 (calibration data)
stimulated node N0 (input during t = 1 to 2)
readout N4 (not recorded)
b
Calibration at recorded nodes
Noisy rates (a.u.), two trials from the same initial state
Trial 1
Trial 2
N6
1.65
1.50
1.35
N8
1.05
0.90
N11
0.75
0.60
0
3
6
9
12
Calibration time (model units; trial 2 shifted by 6)
c
Held-out response at N4
Stimulated minus control; shaded integral gives q
0.6
0
stimulus input at N0 (a.u.)
Rate difference (10⁻⁴ a.u.)
0.75
0.50
0.25
0.00
0
1
2
3
4
5
6
Held-out time (model units)
q = 2.43518 × 10⁻⁴ a.u. × model time